

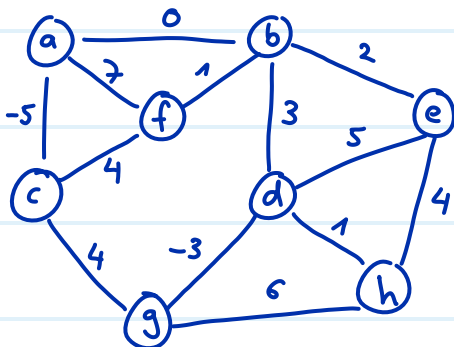
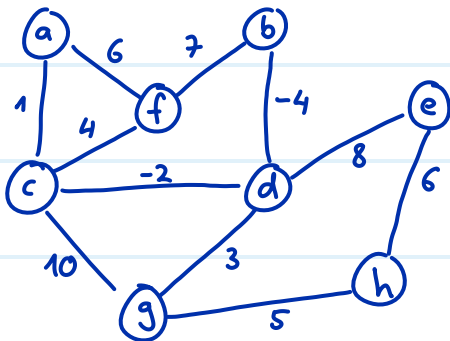
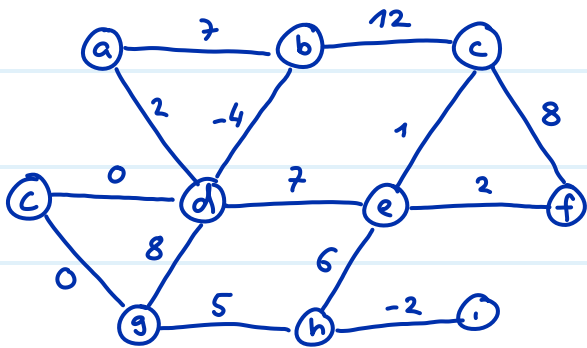
Kruskal

Algorithm 1 KRUSKAL(G)

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1:  $F \leftarrow \emptyset$ 
2: for  $(u, v) \in E$  aufsteigend sortiert nach Gewicht do
3:   if  $u$  und  $v$  in verschiedenen ZHKs von  $F$  then
4:      $F \leftarrow F \cup \{(u, v)\}$ 
5:   end if
6: end for

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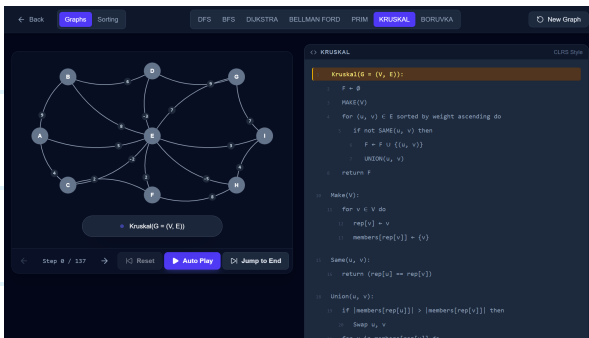
Algorithm 1 KRUSKAL(G) with Union-Find Logic

Data Structures:1: $rep[1 \dots n]$ Invariante · $rep[u] = rep[v] \Leftrightarrow ZHK(u) = ZHK(v)$ 2: $members[1 \dots n]$ $members[u]$ ist eine Liste aller Knoten in $ZHK(u)$ 3: **Make**(V): $O(n)$ 4: **for** $v \in V$ **do**5: $rep[v] \leftarrow v$

jeder Knoten ist eine eigene ZHK

6: $members[rep[v]] \leftarrow \{v\}$ 7: **Same**(u, v): $O(1)$ 8: **return** ($rep[u] == rep[v]$)9: **Union**(u, v): $O(\min\{|ZHK(u)|, |ZHK(v)|\})$ 10: **if** $|members[rep[u]]| > |members[rep[v]]|$ **then**wir wollen möglichst wenige $rep[]$ pointer11: Swap u, v

umhängen

12: **for** x in $members[rep[u]]$ **do**13: $rep[x] \leftarrow rep[v]$ 14: $members[rep[v]] \leftarrow members[rep[v]] \cup \{x\}$ 15: **Kruskal**($G = (V, E)$): $O(\underbrace{m \log m}_{\text{sortieren}} + \underbrace{n \log n}_{\text{union find}}) \leq O(m \log m)$ weil $m \geq n-1$ 16: $F \leftarrow \emptyset$ 17: **MAKE**(V)18: **for** $(u, v) \in E$ sorted by weight ascending **do**19: **if not** **SAME**(u, v) **then**20: $F \leftarrow F \cup \{(u, v)\}$ 21: **UNION**(u, v)22: **return** F 

Tricks for Graph Problems

Problem 1a: The One-Way Street Party

The Scenario: You have a map of a city with N houses and exactly **one** large stadium. The city has many one-way streets (a directed graph). Everyone in the city wants to drive from their house to the stadium for a concert.

You need to calculate the shortest path distance from **every** house to the stadium.

The "Obvious" (Inefficient) Approach: Run Dijkstra's algorithm N times, once from every house node. This is computationally expensive ($O(N \cdot E \log V)$).

Problem 1b: The Mass Evacuation

The Scenario: A storm is approaching. There are N houses in the city (directed graph) and k different storm shelters located at specific nodes. Every household needs to know the distance to the **nearest** shelter. They don't care which one, as long as it is the closest one to their specific house.

Problem 2: The Half-Price Coupon

The Scenario: A traveler wants to fly from City A to City B. They have a map of flights with associated costs. They possess a single "50% off" coupon that can be applied to **exactly one** flight edge in their entire journey. Find the path that minimizes the total cost.

Problem 3: The High-Clearance Truck

The Scenario: You are driving a tall truck from Town A to Town B. The roads have no traffic, but they have old bridges with specific **height limits** (capacities). You want to find a path from A to B such that the **minimum clearance** on that path is maximized (so you can drive the tallest possible truck).

The Challenge: Standard shortest path minimizes the *sum* of weights. Here, we don't care about the sum; we care about the "weakest link" (the lowest bridge).

Problem 4: The Most Reliable Path

The Scenario: You are sending a sensitive message across a computer network. Each cable (edge) has a **reliability probability** P_e (where $0 < P_e \leq 1$). If a message goes through three cables with reliabilities 0.9, 0.9, 0.8, the total success probability is $0.9 \times 0.9 \times 0.8 = 0.648$. Find the path from Start to End with the highest total probability of success.

Problem 5: The Currency Arbitrage

The Scenario: You are given a list of exchange rates between currencies (e.g., 1 USD = 0.85 EUR, 1 EUR = 140 JPY). Is it possible to start with 1 unit of currency, trade through a sequence of other currencies, and end up with *more* than 1 unit of the starting currency?

Problem 6a: The Security Clearance

The Scenario: A spy wants to travel from Safehouse A to Safehouse B . The map includes normal roads and one specific high-security bridge connecting intersection U to V . The bridge (U, V) is locked. To cross it, the spy must first acquire a security pass. The pass is located at a specific pickup point, Node P . Find the shortest path from A to B . (Note: The path might not need the bridge, but if it uses the bridge, P must be visited first).

Problem 6b: The Spy Car

The Scenario: This extends Problem 6a. The spy still needs to go from A to B and requires a pass from P to cross the bridge (U, V) . However, there is now an additional special node C (Car Garage). If the spy visits C , they acquire a fast car. Driving the car allows them to traverse all subsequent roads in **half the time** (edge weights become $W/2$). Find the fastest path from A to B , considering any combination of picking up the pass and/or the car.

Problem 9: The Feuding Families

The Scenario: You are organizing a wedding dinner. You have a list of N guests. You also have a "List of Enemies": pairs of guests who hate each other and will cause a scene if placed at the same table. You have exactly **two** large tables available. Can you assign every guest to one of the two tables such that no two enemies sit together? If so, how?

Problem 10: The Flat-Rate Commute

The Scenario: You have a city with N subway stations. There are K different subway lines. Each line connects a specific list of stations (e.g., Line 1 connects $\{A, C, F\}$). Entering a line costs a flat ticket fee of \$2.00. Once you have entered a line, you can travel to any other station **on that same line** for free. Switching lines requires exiting and paying the ticket fee for the new line. Find the cheapest cost to get from Start to End.

The "Obvious" (Inefficient) Approach: For every subway line, create a "clique": connect every station on the line to every other station on the line with an edge of weight 2.00. If a line has 100 stations, this creates $\approx 10,000$ edges for just that one line!

Problem 12: The Electric Vehicle Trip

The Scenario: You are driving an Electric Vehicle (EV) from City A to City B. The car has a battery capacity of C units. Roads connect cities, and driving from u to v consumes $dist(u, v)$ units of charge. You cannot traverse an edge if you don't have enough charge. Some cities have Charging Stations where you can refill your battery to full capacity (C) instantly. Find the shortest travel time to get from A to B.

what if add a charging time of 5 time units?